The Extension Problem for Orthomodular Observation Algebras

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1 Background

All measurement is finite. An experimenter performs finitely many trials, records finitely many digits, distinguishes finitely many outcomes. Yet the mathematical frameworks we use to organise empirical knowledge — probability theory, quantum mechanics, statistical mechanics — rest on infinite structures: σ-algebras, Hilbert spaces, continuous symmetry groups. The passage from finite observation to infinite structure is not automatic; it requires a commitment that no finite body of evidence can compel.

In probability theory, the locus of this commitment is *countable additivity.* Given a Boolean algebra $\mathcal{E}$ of events and a finitely additive charge $\ell : \mathcal{E} \to [0,1]$ with $\ell(\Omega) = 1$, the condition

$$E_n \downarrow \varnothing \quad \Longrightarrow \quad \ell(E_n) \to 0 \tag{1}$$

— that mass cannot persist along a vanishing sequence of events — is not derivable from any finite collection of observations. De Finetti [8] argued that only finite additivity is empirically grounded; Kolmogorov [21] adopted σ-additivity as an axiom and built modern probability on it. The question of *when* and *why* the passage from one to the other is justified has never been fully resolved.

In quantum theory, the same tension appears in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be simultaneously determined. The closed subspaces of a Hilbert space H form an orthomodular lattice $L(H)$ — a structure in which complementation is well-defined but distribution of meet over join may fail:

$$a \wedge (b \vee c) \;\neq\; (a \wedge b) \vee (a \wedge c) \qquad \text{in general.} \tag{2}$$

A *state* on such a lattice is a function $s : L(H) \to [0,1]$ satisfying

$$s(\mathbf{1}) = 1, \qquad s(a \vee b) = s(a) + s(b) \quad \text{whenever } a \perp b. \tag{3}$$

The non-distributivity means that s is additive only on orthogonal pairs, not on arbitrary finite unions. The classical machinery for extending finitely additive assignments to σ-additive ones — Carathéodory's outer-measure construction, which requires a Boolean algebra of sets — does not apply.

For the specific lattice $L(H)$, this difficulty is resolved by a remarkable rigidity. Gleason's theorem [16] shows that, when $\dim H \geq 3$, every state has the form

$$s(P) \;=\; \mathrm{tr}(\rho\, P) \tag{4}$$

for a unique density operator ρ, and is therefore automatically σ-additive. The geometry of Hilbert space is so constraining that the problem of the infinite simply does not arise — states have no room to misbehave.

But Hilbert space is very special. Orthomodular lattices arise in many other settings — the projection lattices of operator algebras, the logics of quantum systems with superselection rules, and the abstract lattices studied in quantum logic — and for these, no analogue of Gleason's rigidity is known. The question that probability theory and quantum theory share — when does finite coherence force countable behaviour? — remains open in this broader setting, and is the subject of the present note.

The programme "Structure from Observation" [23] approaches this question from a particular direction: rather than beginning with a pre-given space Ω of outcomes and asking which σ-algebras it supports, it begins with the *observations themselves* — a directed system of event algebras $\mathcal{E}_i$ with compatible charges ℓ_i — and asks what the directed structure alone determines.

Paper I answers this for Boolean event algebras. A compatible family of finitely additive charges extends to a σ-additive probability measure if and only if collective exhaustion (CE) holds — condition (1) applied uniformly across the directed system. Two constructions realise the extension:

- **Carathéodory:** assumes CE; produces a measure on $(\Omega, \sigma(\mathcal{C}))$ directly.
- **Stone:** assumes nothing; produces a Baire measure on the Stone space $\mathrm{St}(\mathcal{C})$ unconditionally.

Under discriminability, the two agree. Three questions orient the broader programme:

(1) What must be added to observation to get structure?

(2) Reconstruction without points: what structure is already present in the relational description — contexts, directed refinement — before point-spaces are assumed?

(3) Local-to-global extension: what conditions on the *site* control whether local data extend globally?

Paper I resolves (2) and (3) for Boolean algebras. The present note asks whether any of this survives when the event algebra is orthomodular but not distributive.

Epperson and Zafiris [13] advocate a *relational realist* ontology in which structure is constituted by relations among contexts, not by points in a pre-given space. An orthomodular lattice A is the natural algebraic expression of this: its elements are propositions, its partial order is entailment, and its non-distributivity reflects the impossibility of assigning simultaneous truth values to all propositions. The dual space $S_0(A)$ of McDonald–Bimbó [24] is built from *filters* — consistent collections of propositions — not from points of a pre-existing space. This is question (2) applied to non-Boolean algebras: what structure does the relational description already contain?

In the Boolean case, one might hope that the passage from finite to countable additivity could be understood as a *sheaf condition* — local states gluing to a global σ-additive measure. Biesel [1] shows this hope is disappointed:

- Finitely additive charges form a sheaf for finite covers.
- σ-additive measures form a sheaf for countable covers.
- The step between is the *definition* of σ-additivity, not a structural theorem.

This is the boundary at which the present problem begins.

Paper II [22] introduces three positions on the relationship between observation and reality, each a constraint on where the dual-space measure lives:

Empirical adequacy (EA)

The dual-space measure agrees with all observations.

Probabilistic realism (PR)

The measure descends to a realisation space.

Value-definite realism (VDR)

A global two-valued homomorphism exists.

These are nested: VDR $\Rightarrow$ PR $\Rightarrow$ EA. For Boolean algebras, all three are commensurable — every EA-theory admits a VDR-completion. For OMLs with dim ≥ 3, Gleason's theorem (4) rescues PR, but Kochen–Specker [20] blocks VDR. The extension problem sits at the EA $\rightarrow$ PR transition: can we pass from a state on the algebra (EA) to a σ-additive measure on the dual space (PR)? If so, under what conditions — and is descent then a free choice (as in the Boolean case) or a structural consequence of the algebra and state together?

Van Fraassen's distinction [31] between empirical adequacy and structural commitment, which Paper I makes precise via the Stone construction, thus acquires a sharper form in the OML setting: the Stone route is no longer available, the Carathéodory route is blocked by non-distributivity, and the passage from coherent local data to global probability — question (3) — requires genuinely new ideas.

2 The problem

Let A be an orthomodular lattice (OML) and let $s : A \to [0, 1]$ be a state (3). The McDonald–Bimbó duality [24] associates to A a compact topological space

$$S_0(A) \;=\; \big(F(A),\, \subseteq,\, \perp_A,\, P(A),\, T(S)\big),$$

where $F(A)$ is the set of filters on A and $P(A) \subseteq F(A)$ is the set of prime filters. For each $a \in A$, set

$$h(a) \;=\; \{x \in F(A) : a \in x\}.$$

The map h is an OML isomorphism from A onto the $\perp$-stable clopen sets $\mathrm{CO}(S_0(A))^{\dagger}$. Define a set function μ on these clopens by

$$\mu\big(h(a)\big) \;=\; s(a). \tag{5}$$

Question 2.1 (Extension)**.** Under what conditions on the OML A does μ extend to a σ-additive measure on the Baire (or Borel) σ-algebra of $S_0(A)$?

This is the direct OML analogue of the Stone extension theorem established for Boolean algebras in [23].

In the Boolean case, the extension is automatic. The clopens $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ form a Boolean algebra, the state s restricts to a finitely additive charge on this algebra, and the Carathéodory extension theorem (or Choksi's projective-limit argument [7]) produces the σ-additive extension. Compactness of the Stone space gives σ-subadditivity for free.

In the OML case each of these steps breaks:

(i) $\mathrm{CO}(S_0(A))^{\dagger}$ is an OML, *not* a Boolean algebra.

(ii) The state s is additive on orthogonal pairs only, not on arbitrary finite unions.

(iii) **Carathéodory does not apply** — there is no Boolean algebra of sets on which to run the outer-measure construction.

(iv) Compactness holds (McDonald–Bimbó, Lemma 3.5 [24]), but the σ-subadditivity argument requires orthogonality structure that the topology alone does not provide.

The gap is sharp: "finitely additive on a compact Boolean algebra of clopens" automatically extends; "orthogonally additive on a compact OML of clopen $\perp$-stable sets" does *not*.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (4) resolves the problem completely: every state is induced by a density operator, and the resulting measure is σ-additive on $L(H)$. The extension to $S_0(L(H))$ follows. For general OMLs, no Gleason-type representation is available. Question 2.1 asks for the structural conditions that substitute for the Hilbert-space geometry.

Feature	**Boolean (Paper I)**	**OML (this problem)**
Dual space	$\mathrm{St}(A)$ (ultrafilters)	$S_0(A)$ (all filters)
Distinguished subset	$\mathrm{pure}(\Omega)$, not canonical	$P(A)$, canonical
Clopens	Boolean algebra	OML (non-distributive)
Additivity of s	full (finite)	orthogonal pairs only
Extension	automatic (Carathéodory)	requires Gleason-type result
Realisation constrained?	No	Yes (Kochen–Specker [20])

Table 1: Structural comparison of the extension problem.

Two axes: extension and descent. The passage from a state on A to a measure on $S_0(A)$ splits into two axes that the Boolean case fuses but the OML case separates. Since $S_0(A)$ is a Stone space, its *full* clopen algebra $\mathrm{CO}(S_0(A))$ is Boolean, whereas the state lives only on the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$ — the OML, the image of h. These share only meet; join, complement, and bottom differ.

Extension Extend μ from the $\perp$-stable clopens to a finitely additive charge on the *full* Boolean algebra of clopens — the classical Boolean extension problem [18]. Since h preserves meets, $a \wedge b = 0$ forces $h(a) \cap h(b) = h(0)$, so meet-zero elements of A map to *disjoint* clopens. A charge must therefore satisfy, for every pairwise-meet-zero family $\{a_i\}$, $\sum_i s(a_i) \leq 1$ — a Pitowsky correlation polytope / Bell–Boole inequality system [26]. σ-additivity of the state does not help here.

Compactness

Once extension succeeds, the charge extends to a σ-additive Borel measure automatically. Unconditional.

In the Boolean case (Paper I), the extension axis is *free*: finite additivity on the clopen Boolean algebra plus compactness gives the Stone measure unconditionally — "the first arrow is free." In the OML case, **the extension axis is no longer free.**

The extension obstruction is meet-zero versus orthogonal. Three strengthening conditions on s must be distinguished (Pták–Pulmannová [29], Def. 2):

(1) *State*: additive on *orthogonal* pairs ($a \leq b^\perp$).
(2) *Valuation*: additive on *meet-zero* pairs ($a \wedge b = 0$) — a binary condition, strictly stronger, since in a non-distributive OML meet-zero is coarser than orthogonality.
(3) *Extends to a Boolean charge*: the *n-ary* condition $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family — strictly stronger again.

The extension axis is condition (3). It can fail for *every* state. On the smallest non-distributive OML MO_3 (three $2{\times}2$ blocks pasted at 0 and 1), all six atoms are pairwise meet-zero, so orthoadditivity forces the three complementary pairs to sum to 3 while a charge demands ≤ 1: no state extends. The maximally mixed state $s \equiv \frac{1}{2}$ *is* a valuation (it satisfies the binary condition (2)) yet already fails the n-ary condition (3) at the triple $\{p, q, r\}$, where $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So the obstruction is the gap between meet-zero and orthogonality, not σ-additivity and not non-contextuality.

The concrete/non-concrete boundary is not where the gap closes. A set-representable OMP (P, L) requires only closure under complement and under *disjoint* union (Burešová–Pták [4], Def. 1.1); closure under intersection is *not* required — it would force L to be a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member contained in $A \cap B$, so $a \wedge b \subseteq A \cap B$ with equality iff $A \cap B \in L$. Hence orthogonality ($A \cap B = \varnothing$) implies meet-zero always, but **meet-zero does not imply orthogonality** for incompatible pairs. The smallest witness is MO_2, the concrete logic on $P = \{1, 2, 3, 4\}$ with $L = \{\varnothing, \{1, 2\}, \{3, 4\}, \{1, 3\}, \{2, 4\}, P\}$: there $\{1, 2\} \wedge \{1, 3\} = 0$ while $\{1, 2\} \cap \{1, 3\} = \{1\} \neq \varnothing$. Crucially, MO_3 *itself is concrete* (it carries 2^3 ordering two-valued states, hence is set-representable by Gudder's representation theorem for concrete logics — a concrete logic is exactly an OMP with an order-determining set of two-valued states [28]), so the flagship "meet-zero $\neq$ orthogonal" example above is a concrete logic. The boundary that matters is therefore not concrete versus non-concrete but a *richness* / atomicity condition: meet-zero coincides with orthogonality iff every nonempty $A \cap B$ contains a nonzero member of L (sufficient: all singletons lie in L). This property has no established standard name (its cousins — Jauch–Piron, Tkadlec "regional," Burešová–Pták "point-distinguishing" — are all distinct). (Terminology trap: MO_3's celebrated non-representability is von Neumann *coordinatisation* — failure to be a subspace lattice of a projective geometry — a different notion from set-representability.)

Status of the extension axis. For *rich* concrete OMLs (singletons in L, or the weaker intersection-richness above) meet-zero coincides with orthogonality, and the extension axis reduces to the classical marginal / Pitowsky non-contextuality problem. For *non-rich* concrete OMLs (MO_2, MO_3) and non-concrete OMLs ($L(H)$) the meet-zero/orthogonal gap is non-empty, and the extension axis is strictly stronger and can fail universally. In the *finite* case it is a decidable linear program, not a structurally resistant frontier. What remains open is the *infinite* case — $L(H)$ and the actual σ-additive measure on $S_0(A)$ — to which the finite counterexample does not speak.

Descent. σ-additivity of the state governs descent, not extension. Even once μ extends to a measure $\hat{\mu}$ on $S_0(A)$, the *descent* question — whether $\hat{\mu}$ concentrates on the "physical" points — changes character:

Boolean	Does $\hat{\mu}$ concentrate on pure(Ω)? If and only if the charges are σ-additive. The choice of realisation space Ω is unconstrained.
OML	Does $\hat{\mu}$ concentrate on $P(A)$? Here $P(A)$ is part of the structure, not a free choice. The Kochen–Specker obstruction [20] prevents simultaneous realisation of all filters; the Bub–Clifton theorem [2] shows that the state determines a maximal Boolean subalgebra of definite values.

In the OML setting, the algebra and state *together* constrain which filters are "realised." Descent is not a geometric commitment but a structural consequence.

Remark 2.2 (Decomposition lives on the descent axis)**.** De Simone–Navara [9] decompose OMP states as $s = s_\sigma + s_{\mathrm{wpfa}}$ (a σ-additive part and a weakly purely finitely additive part), the orthomodular analogue of Yosida–Hewitt. By the axis split above, the wpfa component governs the *descent* axis — it is the analogue of the purely finitely additive part ℓ_p in the Boolean case — and *not* the extension axis. A natural-looking conjecture, "s extends iff $s_{\mathrm{wpfa}} = 0$," is therefore mis-targeted: vanishing of s_{wpfa} concerns whether the (already σ-additive) state's measure concentrates on $P(A)$, while extension is blocked by non-distributivity regardless of the wpfa component. The decomposition theory and the extension problem live on different axes.

A caveat compounds this. The McDonald–Bimbó duality [24] is *finitary*: the identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for finite joins but fails in general for countable joins, since finitary OML homomorphisms need not preserve infinite suprema. Making even the descent argument rigorous — the analogue of Paper I's "σ-additivity $\Leftrightarrow$ concentration" — requires σ-completeness and continuity hypotheses the 2023 duality does not supply, and is itself open.

The finitary-to-σ bridge: open, with three named routes blocked. Paper I's descent runs on a *Loomis–Sikorski* engine: σ-additive measures on the Stone space correspond to σ-homomorphisms off the ideal points, which is what makes "σ-additive $\Leftrightarrow$ concentrates on pure(Ω)" go through. The OML descent question is whether that engine has an OML analogue — whether a σ-complete OML admits a Loomis–Sikorski representation (quotient of a σ-tribe of sets by a σ-ideal) or a countable-join- preserving σ-Stone duality. The literature settles only the known *routes*, all of which are blocked.

(i) **RDP / effect-algebra route — blocked.** Loomis–Sikorski holds for σ-complete MV-algebras (Dvurečenskij [11]) and for monotone σ-complete effect algebras *with* the Riesz Decomposition Property (RDP) [12]. But a lattice effect algebra has RDP iff it is an MV-algebra (Riečanová; Paseka), and an OML that is an MV-algebra — all pairs compatible — is Boolean (Kalmbach [19]). So OML + RDP $\Leftrightarrow$ Boolean: the RDP machinery has no non-Boolean OML to act on.

(ii) **MacNeille-completion route — blocked.** Harding [17] shows the MacNeille completion of an OML need not be orthomodular, so one cannot complete to absorb countable joins.

(iii) **σ-Stone-duality route — none exists.** McDonald–Bimbó is irreducibly finitary; Freytes' equational theory of σ-complete OMLs [15] supplies no set/tribe representation.

A concentration statement needs a point space, which a general OML may lack (Greechie stateless lattices are non-concrete). But that is a red herring for the live case: $L(H)$ is non-concrete yet Gleason resolves descent positively, and MO_3 is concrete. The genuinely open class is **concrete, non-Boolean, infinite σ-OMLs**, where the non-concreteness obstruction is absent and the RDP argument blocks only one route. No impossibility theorem is known for this class; the contribution here is the map of dead routes, which sharpens "needs a new idea" into "needs a representation bypassing all three."

3 What is known

In the Boolean case, the Kelley–Vladimirov–Pták characterisation (Fremlin, Theorem 391D [14]) gives a complete algebraic answer:

Theorem 3.1 (KVP). *A Boolean algebra B carries a strictly positive σ-additive measure if and only if B is*

(a) Dedekind σ-complete,

(b) weakly (σ, ∞)-distributive, and

(c) chargeable (carries a strictly positive finitely additive measure).

Each condition is algebraic and non-trivially constraining. For OMLs, the analogues fail or degenerate:

(a) Dedekind σ-completeness makes sense but is very strong.
(b) Weak (σ, ∞)-distributivity has no clean OML analogue — the definition relies on distributivity of $\bigwedge$ over $\bigvee$, which may fail orthomodularly.
(c) Chargeability has no known structural characterisation on OMLs; the existence of a strictly positive finitely additive measure cannot be read off from the lattice structure.

The positive results that do exist for OMLs all require algebraic structure rich enough to approximate Hilbert-space geometry:

Gleason [16] $A = L(H)$, $\dim H \geq 3$. Every state has the form (4); σ-additivity is automatic.

Bunce–Wright [3] A the projection lattice of a JBW-algebra. σ-additivity via operator-algebraic structure.

Chetcuti–Dvurečenskij [6] Lattice effect algebras with completeness conditions close to von Neumann algebras.

The obstructive results show that the Boolean extension machinery cannot be transplanted:

Pták–Pulmannová [28] If every unital subadditive measure on an orthomodular poset is a state, the poset must be Boolean. This is the structural reason no KVP analogue exists for OMLs: conditions strong enough to force σ-additivity collapse the lattice back to a Boolean algebra.

Navara [25] Regularity does *not* force σ-additivity on general orthomodular posets. The Béaver–Cook result is special to von Neumann algebras.

Pták [27] Exotic OML state spaces exist; the obstruction is combinatorial (Greechie pasting), not from the centre.

Recent incremental work confirms the field is active but no one has found the right condition: Voráček–Pták [32] study signed measures on orthomodular posets; Burešová–Pták [5] investigate variations on regularity conditions; De Simone–Navara study Yosida–Hewitt-type decompositions for orthomodular posets.

For the *σ-additivity / descent* side, the gap between "too weak" (allows non-σ-additive states) and "too strong" (collapses to Boolean) appears structurally robust — this is the part where further reading has not produced the answer. (This robustness is about the descent axis; the extension axis is classical, per §2.)

All known exotic OML state spaces are built by **Greechie pasting** [33, 30, 27]. Progress on the descent residue likely requires either:

(a) a new pasting technique that controls σ-additivity, or

(b) an approach that bypasses the state space entirely (e.g., categorical or topos-theoretic [10]).

4 Assessment

Question 2.1, once split into its two axes, is not a single open problem but a settled axis and an open one.

Extension axis: largely settled, and partly degenerate. This is the classical Boolean extension problem [18] / Pitowsky polytope feasibility [26]. Because h preserves meets, it is governed by the gap between meet-zero and orthogonality, which is non-empty whenever the concrete representation is non-rich (and a fortiori in any non-distributive OML). In the finite case it is a decidable linear program; it can fail for *every* state (MO_3). This is not a structurally resistant frontier and does not require a new idea. Two earlier framings of this axis were mistaken. (i) "The Pták–Pulmannová frontier needing a new idea" — Pták–Pulmannová [29] concerns the *supply* of valuations (a unital set forces Boolean), not whether a given state extends, and extension is in any case strictly stronger than the valuation condition. (ii) "Concrete OMLs reduce the axis to Pitowsky non-contextuality because meet = intersection" — false: concreteness requires only disjoint-union closure, and MO_2, MO_3 are concrete logics in which meet-zero $\neq$ orthogonality (§2). The gap closes only under a richness / atomicity hypothesis, not under concreteness.

Descent axis: genuinely open, dead routes mapped. Whether a measure on $S_0(A)$ concentrates on the physical points $P(A)$, and the analogue of Paper I's "σ-additivity $\Leftrightarrow$ concentration," cannot even be stated rigorously without σ-completeness and continuity hypotheses, because the McDonald–Bimbó duality is finitary (the countable-join identity fails). The descent argument runs on a Loomis–Sikorski engine, and the three known routes to an OML analogue are all blocked — the RDP / effect-algebra route (OML + RDP $\Leftrightarrow$ Boolean), the MacNeille completion (Harding), and a σ-Stone duality (none exists; Freytes gives only an equational theory). But *no impossibility theorem* is known for the live class — concrete, non-Boolean, infinite σ-OMLs. This is the live residue, now sharpened from "needs a new idea" to "needs a representation bypassing all three blocked routes."

The infinite extension case. The finite counterexample does not speak to $L(H)$ or to the actual σ-additive measure on $S_0(A)$ for an infinite OML. Whether the meet-zero/orthogonal gap produces the same universal failure infinitely, or whether σ-completeness changes the picture, is open.

Novelty. The formulation via McDonald–Bimbó duality [24] appears to be new; the identification of the extension axis with classical Horn–Tarski / Pitowsky feasibility de-mystifies it. For $L(H)$, Gleason resolves both axes at once.

Status. Extension axis: settled (finite), open (infinite). Descent axis: open, contingent on a rigorous finitary-to-σ bridge. Not a current active lead.

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